



Figure 1: Overall architecture of the proposed LightRAG framework.

includes two key functionalities: i) **Data Indexer** $\varphi(\cdot)$: which involves building a specific data structure $\hat{D}$ based on the external database D . ii) **Data Retriever** $\psi(\cdot)$: The relevant documents are obtained by comparing the query against the indexed data, also denoted as “relevant documents”. By leveraging the information retrieved through $\psi(\cdot)$ along with the initial query q , the generative model $\mathcal{G}(\cdot)$ efficiently produces high-quality, contextually relevant responses.